

Mohammed Ali Safwan Khan

BEng (Hons) Electrical & Electronic Engineering · Aston University

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PROFILE

BEng (Hons) Electrical and Electronic Engineering graduate from Aston University, with hands-on experience in embedded systems, control theory, PCB design, and digital electronics. Student member of the Institution of Engineering and Technology (IET). Seeking a graduate engineering role to apply technical and practical skills within a high-performance engineering team.

EDUCATION

BEng (Hons) Electrical and Electronic Engineering (2.1)

2023 – 2026

Aston University · Birmingham

- IET-accredited integrated degree
- Modules include: Computational Problem Solving, Digital Design (VHDL), Control Systems, Electric Drive Train Control, Embedded Systems
- Skills gained: MATLAB/Simulink, Proteus, Python, C/C++, FreeRTOS, LabVIEW, VHDL

BTEC Level 3 National Extended Diploma in Engineering (D* D* D)

2020 – 2023

Aston University Engineering Academy · Birmingham

- Key units: Computer-Aided Design, Specialist Engineering Project, Electronic Devices and Circuits

GCSEs

2015 – 2020

Holly Lodge High School

- 6 GCSEs (Grades 3–5) including English Language (4) and Mathematics (5)

PROJECTS

Wet Bulb Temperature Monitor

2026

Personal Project

- Designed an ESP32-based outdoor heat-stress monitor (Product V1, Firmware V2) using FreeRTOS dual-core split: sensor/OLED/RGB LED on one core, Wi-Fi/web server on the other
- Integrated BMP280 pressure and TSL2591 light sensors to compute a live WBGT heat-stress index, served via a web dashboard and a diagnostics page

Traction Control System for an RC Vehicle

2025 – 2026

Final Year Dissertation · Aston University

- Designed and built a full TCS from the ground up: custom manufactured PCB, dual-core ESP32 firmware (FreeRTOS), and MATLAB/Simulink control model
- Achieved 50–63% wheel slip reduction at low/mid throttle on wood surface, exceeding the 25% SMART objective

2-Player NeoPixel Snake

2023 – 2024

University Team Project · Aston University

- Team project building a two-player Snake game on a custom NeoPixel display: two 16×8 panels connected in series, driven by an Arduino Mega
- Role: Electronic Procurement Lead — responsible for selecting, sourcing, and testing all components, including LED panels, Arduino Mega, power supply, and joysticks

Audio Amplifier

2023

Individual Project · Aston University

- End-to-end project: amplifier circuit designed from first principles (gain, biasing, frequency response), simulated in Proteus, then physically manufactured and tested

Pencil Case Speaker

2022

Specialist Project · Aston University Engineering Academy

- Designed an enclosure in Fusion 360 and fabricated to house both stationery and a working Kitronics speaker system

TECHNICAL SKILLS

- Firmware — C / C++ · C (AVR) · FreeRTOS · ESP-IDF · Arduino
- Control Systems — PID Design · MATLAB · Simulink · SimPowerSystems
- Hardware Design — PCB Layout · Proteus · PCB Fabrication · Schematic Capture
- Digital Design — VHDL · Logic Design · Digital Circuits
- Instrumentation — LabVIEW · Oscilloscope · Logic Analyser · IMU
- Programming — Python · Object-Oriented Design
- Data Analysis — Matplotlib · CSV Telemetry · NumPy
- Office — Microsoft Word · Excel · PowerPoint · Publisher

INTERESTS & MEMBERSHIPS

IET Student Member · Aston Electronics Society · Aston Islamic Society · Self-study: control systems theory · Full UK driver's licence